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How price sensitive is AI? Benchmarking LLM-generated willingness to pay in pricing analytics

Sebastian Oetzel[1] · **Tobias Maiberger**[2]

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Abstract
Large language models (LLMs) are increasingly used to generate synthetic respondents for marketing research, yet their validity for pricing analytics remains unclear. This study benchmarks LLM-generated willingness-to-pay (WTP) against incentive-compatible real WTP elicited through the Becker-DeGroot-Marschak mechanism across four consumer product studies. We examine whether synthetic WTP reproduces real WTP in terms of means, distributions, price-response functions, profit-maximizing prices, and profit forecasts. The results show that LLM-generated WTP can appear economically plausible while producing materially different pricing implications. For two more hedonic products, synthetic respondents overstate WTP and imply overly optimistic demand and profit outcomes. For two more utilitarian products, results are more heterogeneous, ranging from near mean-level correspondence to substantial WTP underestimation. Across all studies, synthetic and real WTP distributions differ significantly, indicating that mean-level similarity is insufficient for validation. These findings suggest that synthetic respondents may support early-stage pricing exploration but should not be treated as direct substitutes for incentive-compatible real WTP data in price optimization.

Keywords Pricing analytics · Willingness to pay · Large language models · Synthetic respondents · Generative AI · BDM mechanism

Introduction

Measuring consumers' willingness to pay (WTP) is a central task in pricing research, marketing analytics, and managerial practice. Reliable WTP estimates are essential inputs for estimating price response, setting prices, and forecasting profits (Miller et al. 2011; Schlereth et al. 2012). Traditionally, WTP is measured either through direct survey formats or through incentive-compatible mechanisms such as the Becker-DeGroot-Marschak (BDM) procedure (Becker et al. 1964). Incentive-compatible approaches are particularly valuable because they provide a closer approximation to consumers' actual valuations than hypothetical formats (Wertenbroch and Skiera 2002; Miller et al. 2011). At the same time, WTP measurement involves important design choices regarding elicitation procedures and sample selection (Evangelidis et al. 2025). Collecting incentive-compatible WTP data also requires real participants, controlled elicitation procedures, and often real transaction consequences. This makes such data valuable but costly, time-consuming, and difficult to scale in fast-moving pricing contexts.

Against this background, generative artificial intelligence (GenAI), and large language models (LLMs) in particular, are beginning to reshape marketing practice and marketing research (Grewal et al. 2025). As firms increasingly use LLMs for tasks such as customer communication, content creation, analytics, and research support, growing attention has turned to whether LLM-generated synthetic respondents can serve as substitutes for human respondents in marketing research (Arora et al. 2025; Grewal et al. 2025). However, prior research offers mixed evidence on the extent to which LLM-generated synthetic respondents replicate human response patterns across domains and tasks (Sarstedt et al.

✉ Sebastian Oetzel
sebastian.oetzel@w.hs-fulda.de

Tobias Maiberger
tobias.maiberger@h-da.de

[1] Department of Business, Fulda University of Applied Sciences, Leipziger Straße 123, Fulda 36037, Germany

[2] Darmstadt University of Applied Sciences, Schöfferstraße 3, Darmstadt 64295, Germany

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2024). Accordingly, recent work has emphasized that synthetic respondent data should be evaluated against relevant human benchmarks, particularly when used in quantitative studies (Sarstedt et al. 2024).

This validation problem is particularly consequential in pricing analytics. WTP estimates are rarely used only as descriptive measures of average valuation. Instead, they are typically transformed into demand curves, price-response functions, optimal prices, sales expectations, and profit forecasts. Importantly, prior WTP research has shown that mean-level correspondence alone may be insufficient for evaluating an elicitation method because similar mean WTP estimates can coexist with different WTP distributions and managerial pricing implications (Miller et al. 2011). As a result, even small discrepancies between synthetic and real WTP may produce substantially different pricing recommendations. Validation therefore requires more than showing that synthetic and human responses are similar on average. It also requires examining whether synthetic WTP reproduces the distributional structure and decision implications of incentive-compatible real WTP. This raises the question of whether synthetic respondents can provide sufficiently valid WTP estimates for pricing analytics.

Existing research has shown that LLMs can reproduce several marketing-relevant patterns and may be useful for tasks such as perceptual analysis, consumer segmentation, and the simulation of consumer responses (Li et al. 2024, 2025; Arora et al. 2025). In pricing-related settings, Brand et al. (2023) show that GPT-generated responses exhibit economically plausible regularities, including downward-sloping demand curves and state dependence. At the same time, this literature has rarely evaluated synthetic outputs against incentive-compatible human data. This matters because LLM outputs can diverge from human behavior in systematic and sometimes difficult-to-anticipate ways (Gao et al. 2025). It is therefore unclear whether economically plausible LLM-generated WTP is sufficiently accurate for pricing analytics.

Beyond documenting whether LLM-generated and real WTP differ empirically, the validation challenge also raises the question of why such differences should be expected. Real WTP reflects valuations formed under real purchase consequences. In contrast, LLM-generated responses are based on patterns learned from text rather than on product evaluations under real budget constraints. Consequently, synthetic WTP may systematically diverge from incentive-compatible human valuations even when the generated responses appear economically plausible. Understanding this distinction is essential for assessing the suitability of synthetic respondents for pricing analytics.

This article addresses that gap by benchmarking LLM-generated WTP estimates against incentive-compatible real WTP data. We define LLM-generated WTP error as the observed deviation between LLM-generated WTP and BDM-based real WTP. To establish a credible human benchmark, we rely on the BDM mechanism, one of the most widely used incentive-compatible methods for eliciting WTP (Becker et al. 1964; Wertenbroch and Skiera 2002). In the BDM procedure, respondents state the maximum price they are willing to pay and purchase the product if a randomly drawn price does not exceed that amount. Because truthful revelation is incentive-compatible under this mechanism, BDM-based WTP provides a strong benchmark for evaluating LLM-generated WTP.

Our study contributes to marketing analytics research in three ways. First, rather than comparing LLM-generated outputs with hypothetical survey responses, we benchmark synthetic WTP against incentive-compatible WTP elicited from human participants. Second, we contribute conceptually by arguing that the validity of synthetic respondents in pricing analytics should be assessed not only at the response level but also at the decision level. We demonstrate that seemingly plausible synthetic WTP may nevertheless imply different demand structures, optimal prices, and profit forecasts, thereby highlighting the distinction between response plausibility and decision validity. Third, we explore whether the validity of LLM-generated WTP differs across product contexts by comparing two products with stronger hedonic consumption motives and two products with stronger utilitarian consumption motives.

In each study, we elicited real WTP from human participants using the BDM mechanism and generated matched synthetic respondents using the participants' demographic and product-specific information. Across the four studies, we compared real and synthetic WTP on multiple dimensions of validity, including mean differences, confidence intervals, confidence-interval overlap, distributional similarity based on the Kolmogorov-Smirnov test, price-response functions estimated with logit models, and profit-maximizing prices under alternative markup assumptions.

Our findings show that LLM-generated WTP errors are neither negligible nor uniform. For the two more hedonic products, Dubai Chocolate and the Bio Energy Ball, synthetic respondents overstate WTP by 86% and 37%, respectively, and imply substantially more optimistic pricing outcomes. For the two more utilitarian products, the pattern is more mixed. In the Glass Bottle study, synthetic WTP differs from the real benchmark by only about 3% and is not statistically significant, yet synthetic and real WTP still imply significantly different distributions and materially different pricing outcomes. In the Earplugs study, by contrast, synthetic respondents understate WTP by about 27% and imply substantially lower optimal purchase probabilities and profits than the real benchmark.



Taken together, these findings show that LLM-generated WTP can appear economically plausible while still producing pricing errors at the decision level. The study therefore argues that synthetic respondents in pricing analytics should be evaluated not only by the plausibility of their responses but also by the validity of the pricing decisions they support.

The remainder of this article is structured as follows. First, we review the related literature on WTP measurement, synthetic respondents, and the use of LLMs in marketing research and pricing analytics. Next, we describe the design of the four empirical studies, including the BDM-based elicitation of real WTP and the prompt-based generation of synthetic respondents. We then present the empirical results on mean-level error, distributional error, demand estimation, and price optimization across the four studies. Finally, we discuss the implications of the findings for marketing analytics, synthetic respondent validation, and managerial pricing practice, as well as limitations and avenues for future research.

Related literature

Our study builds on three related literature streams. The first concerns established methods for measuring WTP and their role as inputs for pricing analytics. The second concerns the use of LLMs as synthetic respondents in marketing research and economically consequential decision tasks, including the validation of synthetic data and the distinction between response plausibility and decision validity. The third concerns the hedonic-utilitarian distinction in consumer research, which helps characterize the product contexts examined in our studies.

Methods for measuring willingness to pay in pricing analytics

Marketing research has long relied on a range of methods to measure WTP. Miller et al. (2011) classify these methods along two dimensions: whether WTP is elicited in a hypothetical or real context, and whether it is measured directly or indirectly. In hypothetical settings, respondents do not face an actual purchase obligation, whereas in real settings they may be required to buy the product at the end of the elicitation task. Methods of the latter type are incentive-compatible because truthful revelation is payoff-relevant.

Among indirect approaches, choice-based conjoint (CBC) analysis is one of the most widely used methods. Respondents repeatedly choose among product alternatives described by attributes and levels, allowing researchers to infer WTP from observed choice behavior. In its standard form, CBC is hypothetical. When combined with a purchase obligation, it is commonly referred to as incentive-aligned conjoint (ICBC) analysis (Ding et al. 2005). Among direct approaches, open-ended questions ask respondents to state the maximum price they are willing to pay, but typically do so in a hypothetical format. Incentive-compatible direct methods include the Vickrey auction (Vickrey 1961) and the BDM mechanism (Becker et al. 1964). In the BDM mechanism, respondents state their maximum WTP and purchase the product at the randomly drawn price if that price does not exceed their stated WTP. Because truthful reporting is incentive-compatible, BDM is particularly well suited for benchmarking elicited WTP in pricing research.

Prior research provides strong support for the use of incentive-compatible methods. Wertenbroch and Skiera (2002) show that the BDM mechanism is a reliable and valid tool for eliciting consumer WTP in point-of-purchase settings and that it yields significantly lower valuations than hypothetical formats. Miller et al. (2011) compare four common approaches to measuring WTP: the open-ended (OE) format, choice-based conjoint (CBC) analysis, the BDM mechanism, and incentive-aligned choice-based conjoint (ICBC) analysis. They find that BDM and ICBC align most closely with real purchase data. At the same time, their results show that even methods subject to hypothetical bias may still generate useful demand curves and pricing recommendations. Schmidt and Bijmolt (2020), in a meta-analysis comparing hypothetical and real WTP, report an average hypothetical bias of approximately 21%, showing that hypothetical WTP tends to overstate real WTP. Hypothetical bias refers to the difference between hypothetical and incentive-compatible WTP reported by human respondents. In contrast, LLM-generated WTP error refers to the difference between synthetic WTP and incentive-compatible real WTP elicited through the BDM mechanism.

For pricing analytics, WTP estimates are valuable not only as descriptive measures of consumer valuation but also as inputs into demand estimation, price-response modeling, market simulations, and profit optimization. As a result, the validity of WTP data should be assessed not only by comparing average valuations but also by examining whether different data sources imply similar demand curves and pricing recommendations. Our study extends this literature by introducing synthetically generated WTP as an additional source of valuation data and by benchmarking LLM-generated WTP against incentive-compatible BDM data.

Large language models as synthetic respondents in marketing and economic decision tasks

Existing research on the use of LLMs in marketing research is still emerging, but initial studies suggest considerable potential across several domains. Li et al. (2024) show that

LLM-generated brand similarity ratings and product attribute assessments can approximate human survey results in perceptual mapping tasks. Arora et al. (2025) argue more broadly that LLMs can support both qualitative and quantitative marketing research, for example by generating synthetic respondents, moderating interviews, and analyzing unstructured data, while emphasizing the importance of human oversight and prompt design. Similarly, Li et al. (2025) demonstrate the potential of LLMs in consumer segmentation, showing that text-based survey responses can be used to improve clustering accuracy and generate synthetic consumer profiles.

Recent work has also begun to examine LLMs in economically consequential decision tasks. Brand et al. (2023) show that LLM-generated responses can exhibit economically plausible regularities, including downward-sloping demand curves, state dependence, diminishing marginal utility, and reasonable WTP estimates. At the same time, a growing body of work raises doubts about the validity of LLMs as human surrogates. Gao et al. (2025) show that LLM outputs can diverge from human behavior in systematic and sometimes unpredictable ways, depending on prompt language, role framing, and model-specific idiosyncrasies. Dillion et al. (2023) likewise caution against treating LLMs as direct substitutes for human participants because of concerns about representativeness and behavioral fidelity. Relatedly, Goli and Singh (2024) show in the context of intertemporal choice that LLMs can produce preference measures that differ systematically from human benchmarks.

These concerns are particularly relevant when synthetic respondents are used to approximate individual-level preferences or economically consequential choices. More broadly, recent research on synthetic respondents and digital twins suggests that synthetic consumer profiles can only be meaningfully validated when benchmarked against rich human ground-truth data, including demographic, psychological, behavioral, and economic information (Toubia et al. 2025). In line with this view, Sarstedt et al. (2024) argue that synthetic outputs should be benchmarked against real-world data before they are used as substitutes for human respondents. More recently, Petrescu and Krishen (2026) call for systematic research on the boundary conditions of synthetic respondents in marketing, including their sensitivity to prompt design and the conditions under which synthetic data converge with or diverge from human data. This benchmarking logic is especially important in pricing analytics, where synthetic WTP estimates may affect demand estimation, optimal prices, and profit forecasts.

While prior research has primarily examined whether LLM-generated responses resemble those of human participants, comparatively little attention has been paid to why differences between synthetic and human responses should be expected. Incentive-compatible real WTP and LLM-generated WTP arise from fundamentally different response-generation processes. Whereas BDM respondents reveal valuations under binding purchase consequences, LLMs generate contextually appropriate responses based on statistical regularities learned from text rather than economic evaluation under real budget constraints. Consequently, deviations may reflect training information, prompt design, and differences between publicly available information and consumers' private valuations.

This distinction reinforces the need to benchmark synthetic WTP against incentive-compatible human data beyond response-level similarity. Accordingly, we distinguish between response plausibility, referring to the extent to which synthetic responses resemble human responses, and decision validity, referring to the extent to which they lead to comparable managerial pricing decisions.

Taken together, this literature suggests that LLMs can generate plausible and potentially useful marketing data, but that plausibility alone is an insufficient criterion for establishing validity. While Brand et al. (2023) show that LLMs can produce economically plausible WTP estimates, they do not benchmark these estimates against incentive-compatible human data. Our study addresses this gap by comparing LLM-generated synthetic WTP directly with real WTP elicited through the BDM mechanism and by examining not only mean-level similarity but also distributional correspondence, demand estimation, price optimization, and profit implications. In doing so, our study extends the validation of synthetic respondents in pricing analytics from response plausibility to decision validity.

Product contexts: hedonic and utilitarian consumption motives

A well-established distinction in consumer research positions products along a hedonic-utilitarian continuum (Hirschman and Holbrook 1982; Batra and Ahtola 1991). Hedonic consumption is associated with experiential, affective, and symbolic benefits, whereas utilitarian consumption is more strongly related to instrumental and functional performance (Dhar and Wertenbroch 2000; Okada 2005). This distinction is relevant for WTP measurement because consumers may form valuations differently depending on whether a product is primarily evaluated in terms of affective experience or functional usefulness.

For more hedonic products, WTP may be more sensitive to personal taste, situational desire, and affective responses. For more utilitarian products, valuations may depend more strongly on perceived functionality, usage frequency, existing ownership, and external price comparisons. Building on



the conceptual distinction developed in Sect. 2.2, LLM-generated responses are based on statistical regularities learned from large text corpora rather than on valuations revealed under binding economic consequences. Because the information available in the public domain may differ across product contexts, LLMs may approximate valuations differently for more hedonic and more utilitarian products. The direction and magnitude of such differences, however, remain an empirical question.

Against this conceptual background, we use the hedonic-utilitarian distinction to characterize the product contexts examined in our studies. Specifically, we compare two products with stronger hedonic consumption motives, Dubai-style chocolate and a Bio Energy Ball, with two products with stronger utilitarian consumption motives, a reusable glass bottle and earplugs. This comparison allows us to explore whether LLM-generated WTP errors differ across product contexts. However, because the products also differ in category, price level, usage occasion, and available market information, we interpret these differences descriptively rather than as evidence of a causal moderation effect.

Synthesis and research gap

Taken together, the three literature streams reviewed above motivate the present study. Research on WTP measurement identifies incentive-compatible elicitation methods such as the BDM mechanism as the appropriate benchmark for consumer valuation. At the same time, the emerging literature on LLM-generated synthetic respondents highlights both the potential and the limitations of synthetic data but has not yet benchmarked synthetic WTP against incentive-compatible real WTP or examined whether synthetic WTP supports comparable pricing decisions. Finally, the hedonic-utilitarian distinction provides a theoretically motivated product context for examining the boundary conditions under which synthetic WTP may differ from incentive-compatible real WTP. Building on these three literature streams, our study benchmarks LLM-generated WTP against incentive-compatible BDM data. We further evaluate validity not only in terms of response plausibility but also in terms of decision validity for pricing analytics.

Method and data collection

Overview of studies

To examine whether LLM-generated synthetic respondents can provide valid estimates of WTP, we conducted four independent studies comparing synthetic WTP generated by an LLM with real WTP elicited from human participants through an incentive-compatible procedure. In all studies, the human benchmark was obtained through the BDM mechanism, ensuring that participants had an incentive to reveal their true valuations. Across the four studies, a total of 387 participants were recruited on a university campus. For each human participant, we generated a synthetic respondent matched on demographic and product-related characteristics. Throughout the remainder of this paper, we use the term LLM-generated WTP error to refer to the difference between LLM-generated WTP and incentive-compatible real WTP elicited through the BDM mechanism.

All four studies assess WTP for consumer products positioned at different points on the hedonic-utilitarian continuum. For exposition, we present the studies from the more hedonic to the more utilitarian end of that continuum. Study 1 examined a 30 g Dubai-style chocolate bar (hereafter, Dubai Chocolate; $n = 100$). Study 2 examined a 30 g Bio Energy Ball with cocoa and raspberry filling ($n = 91$). Study 3 examined a reusable glass bottle ($n = 100$). Study 4 examined a package of OHROPAX foam earplugs ($n = 96$).

The four products were selected to reflect different positions on the hedonic-utilitarian continuum (Hirschman and Holbrook 1982; Batra and Ahtola 1991). Within this set, the Dubai Chocolate represents the clearest hedonic case, as its value is primarily rooted in sensory pleasure, indulgence, and experiential consumption. The Bio Energy Ball also has hedonic elements, but combines these with more functional associations related to energy and health, placing it closer to the middle of the continuum. The reusable glass bottle likewise occupies an intermediate position: although primarily functional, it may also generate affective and symbolic value through design and sustainability-related meaning. By contrast, the earplugs represent the clearest utilitarian case, as their value is mainly derived from functional problem solving and practical use rather than experiential enjoyment. This product set allows us to examine LLM-generated WTP errors across a range of consumption contexts that differ in hedonic and utilitarian characteristics, price levels, usage occasions, and available market information.

Experimental design for collecting real WTP data

All four studies applied the BDM mechanism, following Wertenbroch and Skiera (2002). The studies were conducted in a standardized point-of-purchase setting on a university campus: participants were approached individually, shown the focal product, and asked on site to state the maximum price they were willing to pay. Under the BDM procedure, a random price was then drawn from a predefined range. If the random price was equal to or below the participant's stated WTP, the participant was required to purchase the product at the drawn price. If the random price exceeded the stated

WTP, no purchase occurred. Because the transaction price is determined by the random draw rather than by the stated WTP, truthful reporting is incentive-compatible under this mechanism.

Before stating their WTP, participants received a standardized explanation of the BDM mechanism and were informed that the purchase decision would be binding. Following Wertenbroch and Skiera (2002), participants were not informed about the specific price range or the distribution of possible random prices before stating their WTP in order to avoid anchoring their valuations. The random-price ranges were defined broadly enough to cover plausible retail prices for the focal products.

The random-price ranges differed by product. For Dubai Chocolate, the random price was drawn uniformly from €0.50 to €6.50 in €0.10 increments. For the Bio Energy Ball, the random price was drawn uniformly from €0.00 to €2.50 in €0.10 increments. For the reusable glass bottle, the random price was drawn uniformly from €0.30 to €14.90 in €0.20 increments. For the OHROPAX earplugs, the random price was drawn uniformly from €0.50 to €5.00 in €0.10 increments. In all studies, participants whose stated WTP was equal to or above the randomly drawn price were required to purchase the product immediately at the drawn price.

For each participant, we recorded the stated WTP. In addition, we collected demographic information, including age, gender, and field of study, as well as product-specific attitudinal variables. These measures were collected both to characterize the participant samples and to construct the synthetic respondent profiles used in the LLM-based elicitation.

Table 1 Product-specific attitudinal measures

Study	Measures	Mean (SD)
1: Dubai Chocolate	How much desire do you currently have for something sweet? (1: "not much"; 5: "a lot")	2.520 (1.329)
	How much do you like Dubai chocolate? (1: "not much"; 5: "a lot")	3.940 (1.062)
2: Bio Energy Ball	How much desire do you currently have for an energy ball? (1: "not much"; 7: "a lot")	2.934 (1.665)
	How attractive do you find the combination of cocoa and raspberry? (1: "not much"; 7: "a lot")	4.758 (1.708)
3: Glass Bottle	How important is sustainability in your everyday life? (1: "not important at all"; 5: "very important")	3.660 (0.867)
	Do you own a reusable bottle? (1: = "yes"; 0: = "no")	0.83 (0.377)
	If you own a reusable bottle: On how many days per week do you use it? (0 to 7 days per week)	5.900 (2.129)
4: Earplugs	How often do you generally use earplugs? (1: "not often"; 5: "very often")	3.083 (1.462)
	How useful do you find this product? (1: "not useful at all"; 5: "very useful")	3.114 (1.520)
	How strong is your current desire to own this product? (1: "not strong at all"; 5: "very strong")	3.104 (1.483)

SD Standard deviation

Table 1 reports the product-specific attitudinal measures collected in each study. In Study 1, participants were asked about their current desire for something sweet and their liking for Dubai Chocolate. In Study 2, participants were asked about their current desire for an energy ball and their evaluation of the cocoa-raspberry flavor combination. In Study 3, the attitudinal measures focused on sustainability and bottle-use behavior, including the importance of sustainability in everyday life, ownership of a reusable bottle, and frequency of use among bottle owners. In Study 4, participants reported prior use of earplugs, perceived usefulness of the product, and current desire to own it.

Prompt design for synthetic data collection

To generate synthetic WTP values, we used a structured prompting approach adapted from Arora et al. (2025), who conceptualize LLM-based synthetic respondents through contextual information, persona construction, and task-specific prompts. We transferred this logic to the present setting by generating a synthetic respondent for each real participant and incorporating the respondent's demographic and product-specific characteristics into study-specific prompt templates. Across the four studies, synthetic WTP was generated using OpenAI GPT models.

For each participant, we created a synthetic respondent matched on demographic characteristics and product-specific attitudinal variables (see Table 1). These participant-specific attributes were inserted into prompts designed to approximate the real elicitation context. Each prompt described the focal product, provided participant-specific information, explained the BDM mechanism, and asked the model to state the maximum price the synthetic respondent would be willing to pay.

Across the four studies, the prompts shared the same general logic but were adapted to the respective product and available participant-level variables. For the products with stronger hedonic consumption motives, the prompts included variables related to current desire and product liking. For the products with stronger utilitarian consumption motives, they included variables related to functionality, ownership, usage, and perceived usefulness.

We applied a common synthetic-respondent logic across all four studies. Rather than optimizing prompts for individual products, we followed the same general procedure for matching synthetic respondents to human participants and constructing the prompts across studies. Study 4 differed

from Studies 1–3 in the OpenAI model and access method used, as detailed below.

The prompt templates differed in some implementation details. For Dubai Chocolate, Bio Energy Ball, and Earplugs, the description of the elicitation procedure included the numerical range from which the random price could be drawn, whereas the Glass Bottle prompt did not include numerical range bounds. The numerical range was included to describe how the random price in the BDM procedure would be determined. However, human participants were not informed about these bounds before stating their WTP. This created a procedural difference between the human and synthetic elicitation procedures that may have influenced synthetic WTP. We therefore conducted a supplementary sensitivity analysis to assess the influence of this prompt feature. Using the same participant profiles, we generated synthetic WTP with and without the numerical range while holding the remaining prompt information constant. The design and results of this analysis are reported in Sect. 4.5. Table 2 illustrates the general prompt structure used for synthetic WTP collection, while the complete prompts for all four studies are reported in the Appendix to facilitate transparency and reproducibility.

Synthetic respondents were generated using the OpenAI environment available for the respective study. Studies 1–3 used the standard ChatGPT-4 interface, whereas Study 4 used GPT-4o via API-based prompting. For Studies 1–3, generation hyperparameters such as temperature could not be manually specified through the standard interface and therefore remained at their platform defaults. For Study 4, no generation hyperparameters were explicitly specified in the API calls, such that the API default settings were used. One synthetic respondent was generated for each human participant, resulting in a one-to-one correspondence between the empirical and synthetic datasets. The model was instructed to report a WTP value in euros using €0.10 increments where specified by the study-specific prompt. The resulting WTP values were recorded as generated by the model and were neither calibrated nor otherwise adjusted to improve correspondence with real WTP. No observations were excluded and no outlier correction or trimming procedures were applied.

Table 2 General prompt structure for synthetic WTP collection

Component	Content
Context	Description of the survey setting, the focal product, relevant product characteristics, and the rules of the BDM mechanism
Persona information	Participant-specific demographic and attitudinal information, including age, gender, field of study, and product-specific variables
Elicitation task	Prompt asking the model to state the maximum price the synthetic respondent would be willing to pay

Results

This section examines how closely LLM-generated WTP aligns with real, incentive-compatible WTP and whether any deviations have meaningful implications for pricing decisions. Across the four studies, we compare real and synthetic WTP along four dimensions: mean differences, distributional differences, demand estimation, and price optimization. Following the logic of Miller et al. (2011), we thus assess synthetic WTP not only in terms of statistical correspondence, but also in terms of its managerial implications for demand estimation and pricing.

Mean-level WTP error

Mean-level WTP error provides a first indication of how closely synthetic WTP approximates real, incentive-compatible WTP. We compare mean WTP across real and synthetic data and examine the corresponding 95% confidence intervals. Following Miller et al. (2011), overlapping confidence intervals provide an initial indication of statistical comparability, whereas non-overlap suggests meaningful differences between methods. We complement this descriptive comparison with two-sample t-tests of mean differences. Table 3 reports the results.

The findings reveal a differentiated pattern across the four studies. For the two products with stronger hedonic consumption motives, synthetic respondents overstate WTP. In Study 1 (Dubai Chocolate), synthetic WTP exceeds real WTP by €1.30, corresponding to a relative difference of approximately +86%, and the difference is statistically significant (t = -9.362, p < 0.001). The confidence intervals do not overlap, indicating limited convergence between the two methods. In Study 2 (Bio Energy Ball), synthetic WTP exceeds real WTP by €0.36, corresponding to a relative difference of approximately +37% (t = -5.515, p < 0.001), again with no overlap between the confidence intervals.

The pattern differs for the two products with stronger utilitarian consumption motives. In Study 3 (Glass Bottle), mean WTP is very similar across methods: synthetic WTP is only €0.19 higher than real WTP, corresponding to a relative difference of about +3%, and the difference is not statistically significant (t = -0.565, p = 0.573). Here, the confidence intervals overlap, suggesting comparable mean-level estimates across the two methods. In Study 4 (Earplugs), however, the direction reverses. Synthetic WTP is €0.82 lower than real WTP, corresponding to a relative difference of about −27%, and the difference is statistically significant (t = 4.866, p < 0.001). The confidence intervals do not overlap, indicating a clear mean-level discrepancy between real and synthetic responses.

Table 3 Mean WTP, standard deviations, and 95% confidence intervals

Study	Method	Mean (SD)	95% CI	Mean difference and t-Test
1: Dubai Chocolate	Real	€1.50 (0.90)	[1.32; 1.68]	Δ = €1.30 Rel. Difference (%): +86% t = -9.362 $p < 0.001$
	Synthetic	€2.80 (1.05)	[2.59; 3.00]	
2: Bio Energy Ball	Real	€0.97 (0.45)	[0.87; 1.06]	Δ = €0.36 Rel. Difference (%): +37% t = -5.515 $p < 0.001$
	Synthetic	€1.32 (0.42)	[1.23; 1.41]	
3: Glass Bottle	Real	€5.46 (3.10)	[4.85; 6.08]	Δ = €0.19 Rel. Difference (%): +3% t = -0.565 $p = 0.573$
	Synthetic	€5.65 (1.06)	[5.44; 5.86]	
4: Earplugs	Real	€2.99 (1.36)	[2.71; 3.26]	Δ = -€0.82 Rel. Difference (%): -27% t = 4.866 $p < 0.001$
	Synthetic	€2.16 (0.94)	[1.97; 2.35]	

Mean difference is calculated as Synthetic − Real. Relative difference is calculated as (Synthetic − Real) / Real × 100. Positive values indicate that synthetic WTP exceeds real WTP, whereas negative values indicate that synthetic WTP is lower than real WTP. Relative differences are calculated using unrounded estimates and may therefore differ slightly from calculations based on the rounded values shown in the table

CI Confidence interval, *SD* Standard deviation

Overall, the results show that LLM-generated WTP error is neither negligible nor uniform across products. While synthetic respondents approximate average valuations in some cases, mean-level similarity is neither consistent across product contexts nor sufficient to establish the validity of synthetic WTP for pricing analytics.

Distributional WTP error

Distributional WTP error captures whether LLM-generated WTP reproduces the shape, dispersion, and location of the real WTP distribution. For pricing decisions, the distribution of WTP is critical because demand functions, price-response relationships, and price differentiation strategies depend on how valuations are distributed across consumers rather than on average WTP alone. We therefore compare not only mean WTP but also the underlying WTP distributions observed in the real and synthetic data. To assess distributional similarity, we use two-sample Kolmogorov-Smirnov (KS) tests. The KS test is well suited to our purposes because it directly evaluates whether the empirical distributions of real and synthetic WTP differ.

The results show that the distributions of real and synthetic WTP differ significantly in all four studies. The largest distributional discrepancy is observed for Study 1 (Dubai Chocolate), where the KS test rejects equality of the two distributions ($D = 0.580$, $p < 0.001$), indicating substantial divergence between the empirical cumulative WTP distributions. Such differences imply that purchase probabilities are estimated differently across price levels, which directly affects demand estimation and price optimization. Although the remaining three studies exhibit smaller KS statistics, distributional equality is likewise rejected for Study 2 (Bio Energy Ball; $D = 0.385$, $p < 0.001$), Study 3 (Glass Bottle; $D = 0.340$, $p < 0.001$), and Study 4 (Earplugs; $D = 0.333$, $p < 0.001$). Notably, the Glass Bottle exhibits one of the smaller KS statistics despite the close agreement in mean WTP, illustrating that mean-level similarity can coexist with distributional differences that translate into different pricing outcomes.

Table 4 Relative WTP differences at the 10th, 50th, and 90th percentiles

Study	10th	50th	90th
1: Dubai Chocolate	+88%	+133%	+72%
2: Bio Energy Ball	+60%	+30%	+20%
3: Glass Bottle	+125%	+14%	-34%
4: Earplugs	-9%	-26%	-27%

Reported values represent the relative difference between synthetic and real WTP at the respective percentile, calculated as (Synthetic − Real) / Real × 100. Positive values indicate higher synthetic WTP, whereas negative values indicate lower synthetic WTP

To better understand where these distributional differences arise, Table 4 reports relative differences between synthetic and real WTP at the 10th, 50th, and 90th percentiles. The results reveal that LLM-generated WTP errors differ systematically across the distribution. For Dubai Chocolate, synthetic WTP exceeded real WTP across all quantiles, with the largest deviation occurring at the median (+133%). For the Bio Energy Ball, overestimation was strongest at the lower end of the distribution, decreasing from 60% at the 10th percentile to 20% at the 90th percentile. In contrast, the Glass Bottle exhibited a markedly different pattern: lower WTP values were substantially overestimated (+125% at the 10th percentile), whereas higher WTP values were underestimated (-34% at the 90th percentile) despite the small difference in mean WTP. For Earplugs, synthetic WTP was consistently lower across the distribution, ranging from −9% at the 10th percentile to approximately −27% at the upper end of the distribution. These findings help explain why products with similar mean WTP can nevertheless yield substantially different demand functions, optimal prices, and expected profits.

Overall, the distributional analysis shows that LLM-generated WTP does not reproduce the empirical WTP distribution in any of the four studies. This is particularly relevant for pricing because price optimization depends on the full distribution of consumer valuations rather than on mean



WTP alone. Thus, mean-level agreement is not sufficient to establish the validity of synthetic WTP for managerial pricing decisions. The following sections examine how these distributional differences affect price-response functions, optimal prices, purchase probabilities, and expected profits.

Price-response functions

To provide concrete managerial implications of synthetic WTP, we next examine whether real and synthetic valuations imply different price-response functions and, consequently, different optimal prices, purchase probabilities, and profit levels. This step is important because differences in WTP estimates can translate into materially different pricing recommendations once they are aggregated into market demand.

To derive price-response functions for both real and synthetic WTP in each study, we transformed the individual WTP observations into price-specific purchase shares on a common price grid. For each price level p, we calculated the proportion of respondents whose stated WTP was at least as high as that price. This yields an empirical survival function that can be interpreted as the share of consumers willing to purchase at price p. We then estimated a binomial logit model of purchase probability as a function of price, $\Pr(buy \mid p) = \exp(\alpha + \beta p)/[1 + \exp(\ \ + ?p)]$, where

Table 5 Logit model fits

Study	Method	α	β
1: Dubai Chocolate	Real	2.762	-1.876
	Synthetic	4.769	-1.707
2: Bio Energy Ball	Real	3.779	-3.920
	Synthetic	5.731	-4.325
3: Glass Bottle	Real	2.958	-0.550
	Synthetic	9.728	-1.721
4: Earplugs	Real	3.851	-1.296
	Synthetic	4.096	-1.896

p denotes the price level, α and β are estimated coefficients, and $\Pr(buy \mid p)$ represents the predicted probability that a consumer purchases the product at price p. Market demand $q(p)$ was then obtained by multiplying the predicted purchase probability by the assumed market volume. The model parameters were estimated by maximum likelihood. Table 5 reports the estimated coefficients, and Fig. 1 plots the resulting price-response functions for real and synthetic data across all four studies.

As expected, the price coefficient β is negative in all estimated models, indicating that higher prices reduce purchase probability. Beyond this common pattern, however, the estimated price-response functions differ meaningfully across real and synthetic data. For the two products with stronger hedonic consumption motives, the synthetic price-response curves are shifted to the right, implying higher purchase

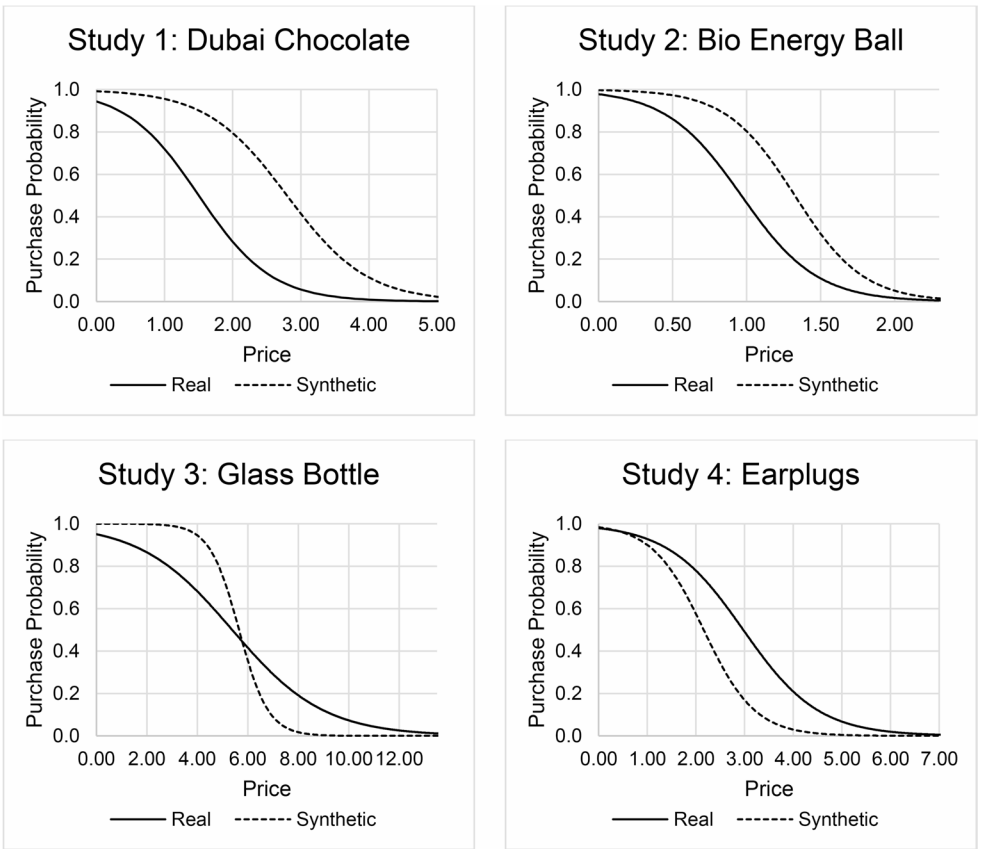


Fig. 1 Price-response functions

probabilities at a given price and therefore higher implied valuations. This pattern is especially pronounced for Study 1 (Dubai Chocolate). For the two products with stronger utilitarian consumption motives, the picture is more heterogeneous. In Study 3 (Glass Bottle), real and synthetic WTP are relatively similar on average, but their price-response curves differ in shape. This shows that similar mean WTP does not necessarily imply a similar demand structure. In Study 4 (Earplugs), the synthetic price-response curve lies below the real price-response curve over much of the relevant price range, implying lower predicted purchase probabilities and lower synthetic valuations.

Taken together, the price-response analysis shows that differences between synthetic and real WTP extend beyond average levels and often imply systematically different market responses to price. This distinction is managerially important because pricing decisions depend on the shape and position of the price-response function, not only on average WTP.

Pricing implications: optimal prices, purchase probabilities, and profits

Based on the estimated price-response functions, we next derive the profit-maximizing price. Expected profit is given by $\Pi(p) = (p - c) \cdot Pr(buy \mid p) \cdot mv$, where c denotes variable cost and mv denotes market volume. We set market volume to 9,000, corresponding to the number of students on campus. Because mv is constant, it scales profits but does not affect the optimal price. Variable costs are derived from observed retail prices under alternative assumed retailer markups of 20%, 30%, 40%, and 50%. Specifically, variable cost is calculated as $c = r/(1 + m)$, where r denotes the observed retail price and m denotes the assumed retailer markup. As a result, higher assumed markups imply lower variable costs and higher unit profits at a given price. The observed retail prices are €2.99 for Dubai Chocolate, €1.50 for the Bio Energy Ball, €6.99 for the glass bottle, and €2.99 for the earplugs. For each markup assumption, we compute the profit-maximizing price and the corresponding predicted purchase probability and profit. To assess uncertainty, we estimate 95% confidence intervals for optimal prices using percentile bootstrap confidence intervals and examine whether the intervals from real and synthetic data overlap. Table 6 reports the results for the two products with stronger hedonic consumption motives, and Table 7 reports the results for the two products with stronger utilitarian consumption motives.

The results show that differences between real and synthetic WTP translate into economically meaningful differences in optimal prices, purchase probabilities, and profit forecasts. For the two products with stronger hedonic consumption motives, synthetic data imply consistently higher optimal prices than the real benchmark. Confidence intervals overlap for Dubai Chocolate at the 20% and 30% markup levels and for Bio Energy Ball at all four markup levels, although the synthetic estimates remain higher throughout. In Study 1, Dubai Chocolate, the absolute price difference ranges from €0.27 to €0.40 across the four markup assumptions, corresponding to an average relative difference of 12.17%. The differences in implied purchase probabilities and profits are substantially larger: on average, synthetic data imply a 375.17% higher optimal purchase probability and 646.82% higher expected profit than the real benchmark. In Study 2, Bio Energy Ball, the absolute price difference ranges from €0.03 to €0.08, corresponding to an average relative difference of 3.95%. Although these price differences are modest in absolute terms, synthetic data still imply a 139.89% higher optimal purchase probability and 181.98% higher expected profit than the real benchmark on average.

The two products with stronger utilitarian consumption motives show a different pattern. In Study 3, Glass Bottle, real data imply consistently higher optimal prices than synthetic data, with absolute differences ranging from €1.42 to €1.53 and an average relative difference of -19.60%. At the same time, the pattern for optimal purchase probability is mixed. At the 20% markup level, the synthetic estimate is slightly below the real benchmark, but at higher markup levels it exceeds it. Averaged across markup assumptions, synthetic data imply a 38.07% higher optimal purchase probability, while expected profit is 47.58% lower than under the real benchmark. This finding reinforces the earlier result that apparently similar mean WTP can still produce materially different demand structures and pricing outcomes. In Study 4, Earplugs, real data again imply consistently higher optimal prices than synthetic data, with absolute differences between €0.49 and €0.57 and an average relative difference of -15.57%. Here, synthetic data also understate both optimal purchase probability and expected profit throughout. On average, synthetic data imply a 44.11% lower optimal purchase probability and a 68.92% lower expected profit than the real benchmark.

Overall, the optimal-price results reveal a consistent directional pattern across the four studies. For both products with stronger hedonic consumption motives, synthetic WTP implies higher profit-maximizing prices than the real benchmark across all markup assumptions. For both products with stronger utilitarian consumption motives, synthetic WTP implies lower profit-maximizing prices throughout. Given the small number of products, this pattern should be interpreted descriptively rather than as evidence of a general moderating effect of consumption motive. Importantly, the Glass Bottle study demonstrates that even close



Table 6 Pricing implications for products with stronger hedonic consumption motives

Study	Method	Assumed Markup	Optimal Price	95% CI	Opt.Price Difference (absolute / relative %)	Optimal Purchase Probability	Opt. Purch. Probability Difference (absolute / relative %)	Optimal Profits	Opt. Profit Difference (absolute / relative %)
1: Dubai Chocolate	Real	20%	€3.05	[2.86; 3.25]	€0.27 / 8.85%	0.049	0.240 / 487.83%	€247.49	€1,910.85 / 772.09%
	Synthetic	20%	€3.32	[3.21; 3.42]		0.290		€2,158.34	
	Real	30%	€2.87	[2.67; 3.08]	€0.32 / 11.15%	0.068	0.269 / 398.08%	€347.29	€2,353.59 / 677.70%
	Synthetic	30%	€3.19	[3.07; 3.29]		0.337		€2,700.88	
	Real	40%	€2.72	[2.52; 2.93]	€0.36 / 13.24%	0.088	0.293 / 333.34%	€461.55	€2,770.83 / 600.34%
	Synthetic	40%	€3.08	[2.96; 3.19]		0.380		€3,232.38	
	Real	50%	€2.59	[2.38; 2.81]	€0.40 / 15.44%	0.109	0.308 / 281.44%	€587.28	€3,154.58 / 537.15%
	Synthetic	50%	€2.99	[2.87; 3.11]		0.417		€3,741.86	
Average across markup levels					€0.34 / 12.17%		0.278 / 375.17%		€2,547.46 / 646.82%
2: Bio Energy Ball	Real	20%	€1.53	[1.48; 1.59]	€0.03 / 1.96%	0.098	0.168 / 170.86%	€247.27	€494.23 / 199.88%
	Synthetic	20%	€1.56	[1.52; 1.61]		0.266		€741.50	
	Real	30%	€1.45	[1.39; 1.51]	€0.05 / 3.45%	0.130	0.190 / 146.46%	€345.38	€649.57 / 188.07%
	Synthetic	30%	€1.50	[1.45; 1.54]		0.319		€994.95	
	Real	40%	€1.38	[1.32; 1.44]	€0.06 / 4.35%	0.164	0.214 / 130.91%	€454.87	€799.69 / 175.81%
	Synthetic	40%	€1.44	[1.39; 1.49]		0.378		€1,254.56	
	Real	50%	€1.32	[1.26; 1.38]	€0.08 / 6.06%	0.199	0.221 / 111.32%	€571.95	€938.83 / 164.15%
	Synthetic	50%	€1.40	[1.35; 1.45]		0.420		€1,510.77	
Average across markup levels					€0.06 / 3.95%		0.198 / 139.89%		€720.58 / 181.98%

For each outcome, the first entry in the difference column reports the absolute difference between the synthetic and real estimates. The second entry reports the relative difference, calculated as (Synthetic − Real) / Real × 100. Absolute differences are reported in euros for prices and profits and in probability points for purchase probabilities. Positive relative differences indicate higher synthetic estimates; negative values indicate lower synthetic estimates. Relative differences are calculated using unrounded estimates and may therefore differ slightly from calculations based on the rounded values shown in the table

correspondence in mean WTP does not guarantee decision validity, as similar average valuations translate into substantially different optimal prices.

Sensitivity to price-range information in the prompt

To assess whether numerical price-range information influenced synthetic WTP, we conducted a supplementary sensitivity analysis. In the original synthetic elicitation, the BDM procedure was represented in a self-contained text-based prompt that specified the random-price mechanism and the response format. For Dubai Chocolate, Bio Energy Ball, and Earplugs, this specification also included numerical price-range information to provide the model with an explicit elicitation context. This created a procedural difference from the human BDM elicitation, as human participants were not informed about these bounds before stating their WTP. To examine the sensitivity of synthetic WTP to this design choice, we generated new synthetic WTP for the same participant profiles under two conditions: a Range condition, which replicated the original prompt specification, and a No-Range condition, which omitted the numerical range while keeping the remaining prompt information unchanged. The Glass Bottle study was excluded because its original synthetic prompt did not contain numerical range bounds. Table 8 compares the two conditions in terms of mean WTP, distributional correspondence, and optimal-price accuracy, with optimal-price differences averaged across the four markup assumptions used in the main analysis.

As a first step, the Range condition provides a replication check of the original synthetic results. The direction of the original WTP differences was reproduced in all three studies: synthetic WTP remained higher than real WTP for Dubai Chocolate and the Bio Energy Ball and lower for Earplugs. The pricing implications were also similar. The average absolute relative optimal-price differences in the original analyses were 12.2%, 4.0%, and 15.6% for Dubai Chocolate, Bio Energy Ball, and Earplugs, respectively, compared with 9.6%, 6.8%, and 16.9% in the newly generated Range condition. Thus, the main pricing implications are not specific to a single synthetic draw.

Against this replicated baseline, removing the numerical range produced substantial but product-specific changes. For Dubai Chocolate, the absolute relative mean difference decreased from 99.1% to 38.2%, and the KS statistic decreased from $D = 0.73$ to $D = 0.46$. For Earplugs, the corresponding values decreased from 31.4% to 1.7% and from

Table 7 Pricing implications for products with stronger utilitarian consumption motives

Study	Method	Assumed Markup	Optimal Price	95% CI	Opt.Price Difference (absolute / relative %)	Optimal Purchase Probability	Opt. Purch. Probability Difference (absolute / relative %)	Optimal Profits	Opt. Profit Difference (absolute / relative %)
3: Glass Bottle	Real	20%	€8.06	[7.71; 8.37]	€1.53 / -18.98%	0.186	0.005 / -2.79%	€3,745.82	€2,597.24 / -69.34%
	Synthetic	20%	€6.53	[6.40; 6.66]		0.181		€1,148.58	
	Real	30%	€7.70	[7.33; 8.04]	€1.51 / -19.61%	0.218	0.066 / 30.25%	€4,560.03	€2,481.26 / -54.41%
	Synthetic	30%	€6.19	[6.05; 6.33]		0.284		€2,078.76	
	Real	40%	€7.41	[7.03; 7.75]	€1.48 / -19.97%	0.247	0.136 / 55.35%	€5,362.88	€2,132.71 / -39.77%
	Synthetic	40%	€5.93	[5.79; 6.07]		0.383		€3,230.17	
	Real	50%	€7.16	[6.79; 7.51]	€1.42 / -19.83%	0.273	0.190 / 69.48%	€6,141.16	€1,644.85 / -26.78%
	Synthetic	50%	€5.74	[5.60; 5.88]		0.463		€4,496.31	
Average across markup levels					€1.49 / -19.60%		0.099 / 38.07%		€2,214.02 / -47.58%
4: Earplugs	Real	20%	€3.60	[3.45; 3.76]	€0.49 / -13.61%	0.307	0.166 / -53.89%	€3,064.93	€2,276.44 / -74.27%
	Synthetic	20%	€3.11	[3.02; 3.18]		0.142		€788.49	
	Real	30%	€3.47	[3.31; 3.63]	€0.52 / -14.99%	0.344	0.161 / -46.92%	€3,624.68	€2,555.71 / -70.51%
	Synthetic	30%	€2.95	[2.86; 3.03]		0.183		€1,068.98	
	Real	40%	€3.37	[3.20; 3.53]	€0.55 / -16.32%	0.374	0.152 / -40.53%	€4,154.95	€2,784.97 / -67.03%
	Synthetic	40%	€2.82	[2.72; 2.90]		0.222		€1,369.97	
	Real	50%	€3.28	[3.11; 3.45]	€0.57 / -17.38%	0.402	0.141 / -35.13%	€4,651.76	€2,970.91 / -63.87%
	Synthetic	50%	€2.71	[2.62; 2.79]		0.261		€1,680.85	
Average across markup levels					€0.53 / -15.57%		0.155 / -44.11%		€2,647.01 / -68.92%

For each outcome, the first entry in the difference column reports the absolute difference between the synthetic and real estimates. The second entry reports the relative difference, calculated as (Synthetic − Real) / Real × 100. Absolute differences are reported in euros for prices and profits and in probability points for purchase probabilities. Positive relative differences indicate higher synthetic estimates; negative values indicate lower synthetic estimates. Relative differences are calculated using unrounded estimates and may therefore differ slightly from calculations based on the rounded values shown in the table

Table 8 Prompt-ablation results

Study	Absolute Relative Mean Difference (%)		KS D		Average Absolute Relative Optimal-Price Difference (%)	
	Range	No-Range	Range	No-Range	Range	No-Range
1: Dubai Chocolate	99.1%	38.2%	0.73	0.46	9.6%	6.5%
2: Bio Energy Ball	53.9%	108.3%	0.60	0.67	6.8%	35.6%
4: Earplugs	31.4%	1.7%	0.39	0.12	16.9%	5.5%

Absolute relative differences are calculated as |Synthetic − Real| / Real × 100. The absolute relative optimal-price difference is averaged across the four markup scenarios

D = 0.39 to D = 0.12. For the Bio Energy Ball, however, correspondence with real WTP became worse: the absolute relative mean difference increased from 53.9% to 108.3%, and the KS statistic increased from D = 0.60 to D = 0.67.

These changes also affected the resulting optimal prices. Removing the numerical range reduced the average absolute relative optimal-price difference from 9.6% to 6.5% for Dubai Chocolate and from 16.9% to 5.5% for Earplugs, but increased it from 6.8% to 35.6% for the Bio Energy Ball. This directional pattern was consistent across all four markup assumptions.

Overall, the ablation analysis yields two conclusions. First, the Range condition broadly reproduces the original pricing implications. Second, removing the numerical range can substantially change synthetic WTP and the resulting optimal prices, but not in a consistent direction across products. Thus, correcting this procedural difference does not provide a general solution for aligning synthetic with real WTP. Instead, the results highlight the need to assess prompt robustness and evaluate synthetic WTP at the level of the pricing decisions derived from it.

Discussion and conclusion

This study examined whether LLM-generated synthetic respondents can recover consumer WTP with sufficient accuracy to support managerial pricing decisions. Across four empirical studies, we benchmarked LLM-generated WTP against incentive-compatible real WTP elicited using



the BDM mechanism. The findings indicate that uncalibrated synthetic WTP should not be treated as a direct substitute for incentive-compatible real WTP in pricing analytics.

The distinction between human and synthetic response generation helps explain why the two WTP measures should not necessarily coincide. Whereas incentive-compatible real WTP reflects valuations formed under real purchase consequences, LLMs generate WTP responses from learned language patterns without facing those economic consequences. Consequently, the plausibility of an LLM-generated WTP response does not imply that it reproduces the economic valuation that a human participant would reveal under incentive-compatible conditions.

A central finding is that LLM-generated WTP errors are neither negligible nor uniform. For the two products with stronger hedonic consumption motives, synthetic respondents overstated WTP by 86% for Dubai Chocolate and 37% for the Bio Energy Ball. These mean-level errors translated into average expected-profit overestimates of 647% and 182%, respectively. Compared with the average hypothetical WTP overstatement of 21% reported in Schmidt and Bijmolt's (2020) meta-analysis, these discrepancies are substantially larger. For the two products with stronger utilitarian consumption motives, the pattern was more heterogeneous. In the Glass Bottle study, mean WTP differed by only 3% and was not statistically significant, yet expected profit was still understated by 48% on average. In the Earplugs study, the direction reversed, with synthetic WTP falling 27% below real WTP and implying 69% lower expected profit. Across all four studies, distributional tests reject equality of real and synthetic WTP distributions, confirming that mean-level alignment is not a sufficient validity criterion for pricing purposes (Miller et al. 2011). The Glass Bottle result is particularly instructive: apparent mean-level correspondence can coexist with substantially different demand structures and pricing outcomes. At the decision level, the optimal-price results reveal a consistent directional pattern within the four studies: synthetic WTP implies higher optimal prices for both products with stronger hedonic consumption motives and lower optimal prices for both products with stronger utilitarian consumption motives.

This study contributes to the literature in three ways. First, it extends the methodological literature on WTP measurement by introducing LLM-generated WTP as a distinct source of valuation data and benchmarking it against incentive-compatible real WTP rather than against hypothetical survey responses. Whereas Schmidt and Bijmolt (2020) report an average hypothetical WTP overstatement of 21%, our results show that LLM-generated WTP errors can be larger in magnitude and less predictable in direction, pointing to a methodological challenge that differs from conventional hypothetical WTP overstatement. Second, the study contributes to the emerging literature on LLMs in marketing research by showing that response plausibility is not equivalent to decision validity. While Brand et al. (2023) demonstrate that LLMs can produce economically plausible regularities such as downward-sloping demand curves, our results show that such regularities can coexist with distributional misalignment and materially different pricing outcomes. Third, the study contributes to pricing analytics by demonstrating that the validity of synthetic WTP should be evaluated at the decision level rather than solely at the response level. WTP estimates are transformed into price-response functions, optimal prices, purchase probabilities, and expected-profit forecasts. As a result, even modest or statistically insignificant mean-level differences may produce economically meaningful differences in pricing recommendations.

The findings also suggest a practical approach for managers considering synthetic WTP in pricing analytics. Synthetic respondents may be useful for exploratory pricing analyses and scenario generation, where speed and low marginal cost are more important than precise price optimization. Before synthetic WTP is used for consequential pricing decisions, however, managers should assess the robustness of the resulting recommendations. Our sensitivity analysis illustrates one such diagnostic: varying the inclusion of numerical price-range information affected synthetic WTP and the resulting optimal prices, although the direction and magnitude of these effects differed across products. More generally, managers can assess the robustness of synthetic WTP across alternative defensible prompt specifications before relying on it for decision making. Prompt stability alone, however, does not establish decision validity, as synthetic estimates may still differ from real WTP and the pricing decisions derived from it. Human validation therefore remains particularly important for consequential pricing decisions. Such validation need not require a complete human benchmark for every synthetic application; rather, firms may use human data to assess whether a synthetic approach performs sufficiently well within a defined application context. Our results caution against assuming that such validation transfers automatically across products or prompt specifications. Similarly, the heterogeneous direction and magnitude of the observed WTP differences do not support a general calibration factor across products.

This study has several limitations that provide opportunities for future research. First, the analysis focuses on four relatively inexpensive, low-involvement consumer goods. Future research should examine whether the magnitude and direction of LLM-generated WTP error generalize to higher-involvement products, services, luxury goods, and other market settings. Second, the human benchmark is



based on student samples from a single German university. Replication across broader consumer populations and cultural contexts is therefore needed. Third, numerical price-range information was included in three of the synthetic prompts as part of the explanation of the BDM procedure, whereas human participants were not informed about these bounds before stating their WTP. We therefore examined the sensitivity of the synthetic results to this prompt feature. The newly generated Range condition broadly reproduced the direction and pricing implications of the original synthetic results, while removing the numerical range affected the results differently across products. Future research could examine the role of numerical information and other prompt-design choices more systematically across products and elicitation settings. Future research should therefore examine prompt sensitivity more systematically across products and elicitation designs. Fourth, the study relies on models from the OpenAI GPT family, with Studies 1–3 using ChatGPT-4 and Study 4 using GPT-4o. Differences across studies may therefore partly reflect model version and implementation rather than product context alone. Future research should test whether the observed patterns generalize across LLMs, model versions, and generation settings.

Overall, LLM-generated synthetic respondents should not yet be viewed as direct substitutes for incentive-compatible real WTP data in pricing research. Their current value appears strongest for exploratory applications in which speed, scale, and low marginal cost are important. For consequential pricing decisions, however, synthetic estimates require greater scrutiny because even plausible or mean-aligned WTP estimates can imply materially different pricing outcomes. More broadly, our findings suggest that the relevant criterion for evaluating synthetic respondents in pricing research is not response plausibility alone, but decision validity.

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Declarations

Conflicts of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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Sebastian Oetzel is Professor of Business Administration with a specialization in Marketing at Fulda University of Applied Sciences, Germany. His research focuses on marketing analytics, pricing, consumer decision-making, user-generated content, and the use of artificial intelligence and synthetic respondents in marketing research.

Tobias Maiberger is Professor of Digital Transformation at Darmstadt University of Applied Sciences, Germany. His research focuses on online reviews, social media, chatbot interactions, and online advertising.